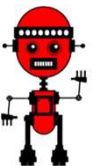
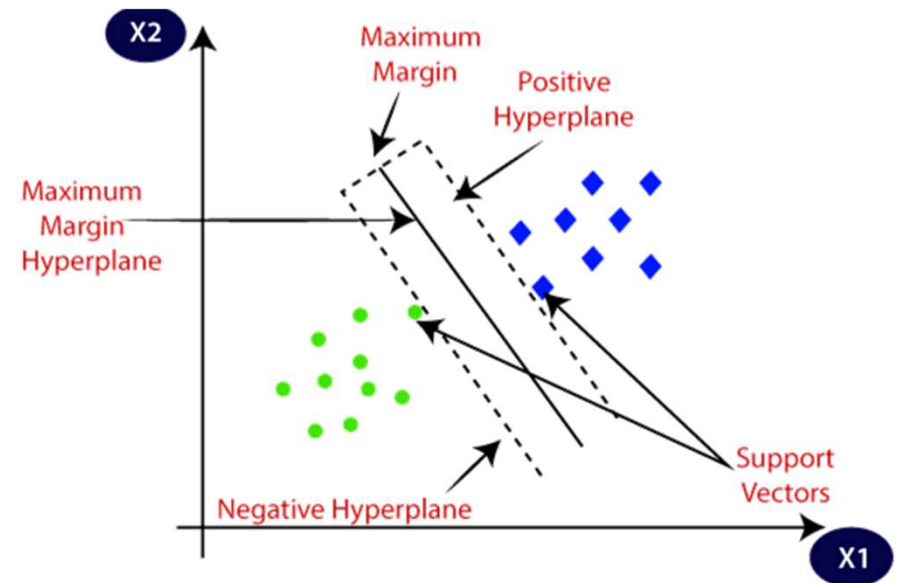


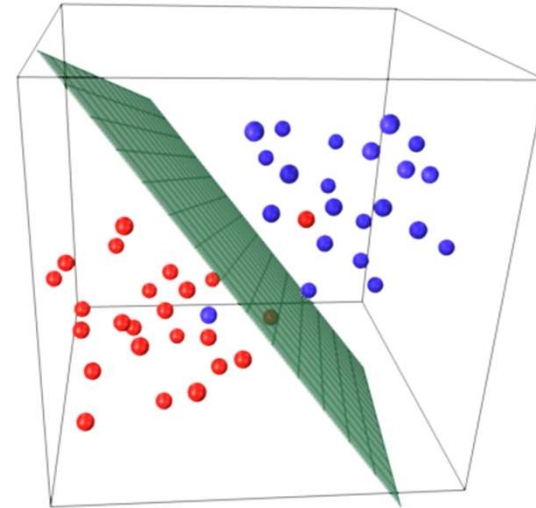
Concept of SVM

- Comes under Supervised machine learning algorithm.
- Can be used for both classification or regression challenges.
- The goal of the SVM algorithm is to create the best line or decision boundary that can segregate n-dimensional space into classes.



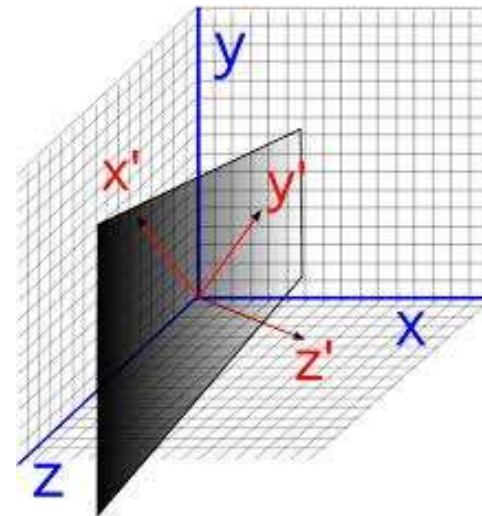
N - dimensional space ?

- The dimension of the hyperplane depends upon the number of features.
- If the number of input features is two, then the hyperplane is just a line.
- If the number of input features are three, then the hyperplane would be 2-D plane.



HYPERPLANE ?

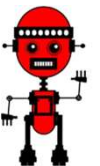
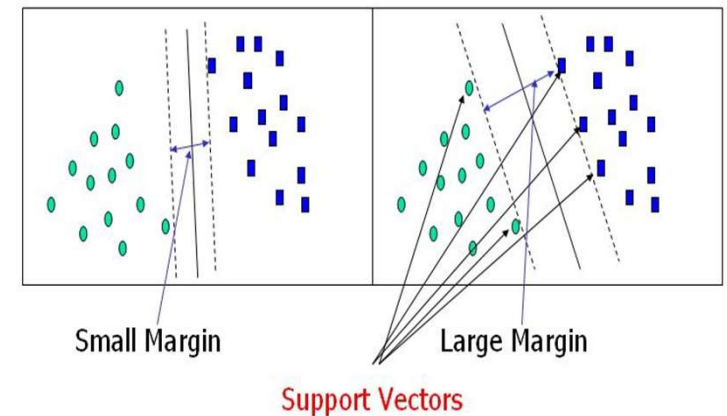
- Hyperplanes are decision boundaries that help classify the data points.
- We always create a hyperplane that has a maximum margin, which means the maximum distance between the data points.



Terminology of SVM

Support Vector And Margin

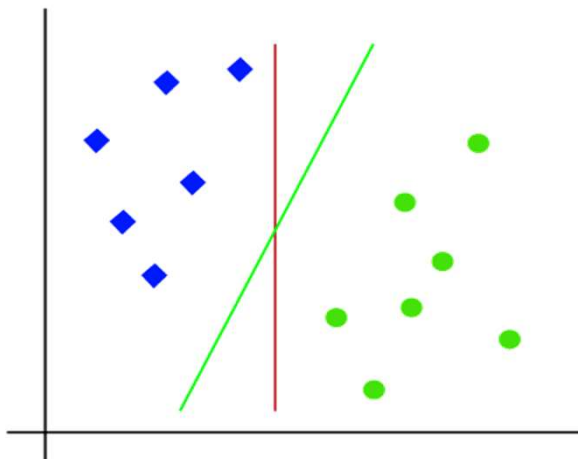
- Support vectors are data points that are closer to the hyperplane and influence the position and orientation of the hyperplane.
- Margin is the perpendicular distance between the closest data points (support vector) and the Hyperplane (on both sides) .



Types of SVM

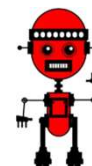
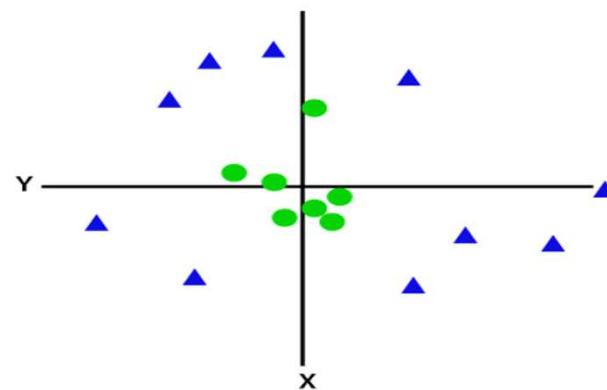
Linear SVM:

- Used for linearly separable data.
- Dataset which can be classified into two classes by using a single straight line, such data is termed as linearly separable data.



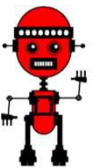
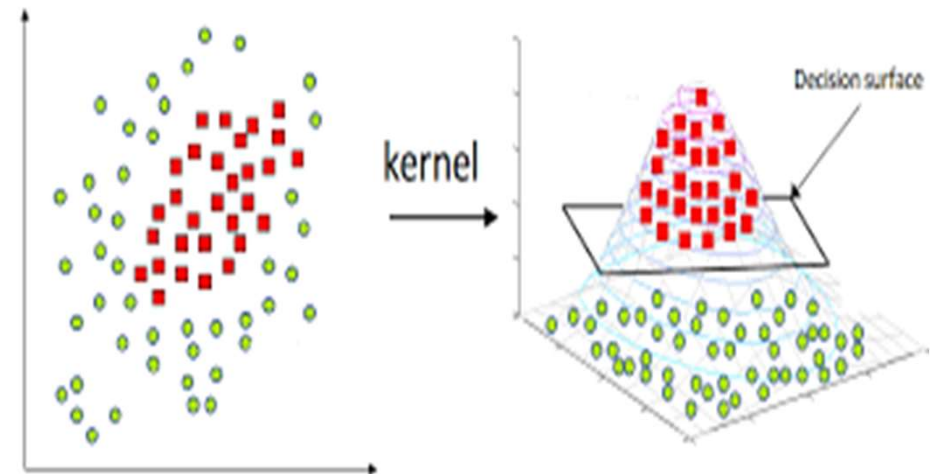
Non-linear SVM:

- Used for non-linearly separated data.
- Dataset which cannot be classified by using a straight line, such data is termed as non-linear data.



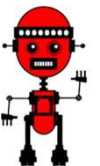
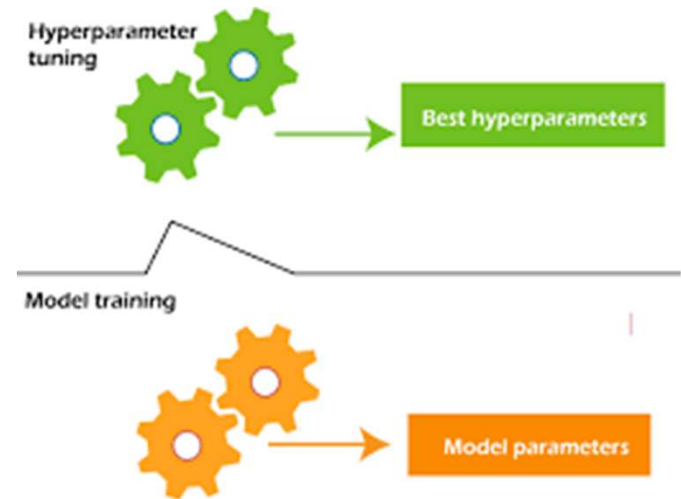
KERNEL FUNCTION

- The SVM kernel is a function that takes low dimensional input space and transforms it to a higher dimensional space.
- Converts not separable problem to separable problem.
- Useful in non-linear separation problem.



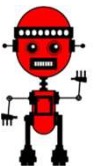
Hyperparameter tuning

- **Hyperparameters –**
 - Parameters that are explicitly defined by the user to control the learning process.
 - Used to improve the learning of the model, and their values are set before starting the learning process of the model.
- The process of selecting the best hyperparameters to use is known as hyperparameter tuning.
- Tuning process is also known as hyperparameter optimization.



Grid Search CV

- An optimization algorithm that allows us to select the best parameters (hyperparameter combination) to optimize result.
- It accepts the possible value for every parameter and executes the code in order to try out each and every possible combination outputs the result for the combinations and outputs the combination having the best accuracy.



USE CASES

- Face detection
- Text and hypertext categorization
- Classification of images
- Bioinformatics
- Handwriting recognition
- Protein fold and remote homology detection.
- Generalized predictive control(GPC)

